INTRODUCTION TO COMPUTERS

1. Types of Computer

Analog

Digital

Hybrid

Analog

Continuous Data

Example

Thermometer

Speedometer

Digital

0 aur 1

Example

Laptop

Computer

Calculator

Phone

Hybrid

Analog + Digital

Example

Hospital Machine

Weather Forecast System

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2. Components of Computer

Input

↓

CPU

↓

Output

CPU ke andar

Control Unit

ALU

RAM

Input Device

Keyboard

Mouse

Scanner

Microphone

Output Device

Monitor

Printer

Speaker

Projector

ALU

Calculation

Decision

Comparison

Control Unit

Sabko Control karta hai.

RAM

Temporary Memory.

Computer OFF

↓

Data Delete.

Hard Disk / SSD

Permanent Memory.

Computer OFF

↓

Data Safe.

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3. Hardware vs Software

Hardware

Jo Touch kar sakte ho.

Examples

Keyboard

Monitor

CPU

RAM

Printer

Software

Jo Touch nahi kar sakte.

Examples

Windows

Python

Chrome

MS Word

VS Code

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4. Operating System

Definition

Operating System

User aur Hardware ke beech Interface hai.

Examples

Windows

Linux

macOS

Android

Functions

Memory Management

↓

File Management

↓

Device Management

↓

Security

↓

Program Execution

↓

User Interface

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5. Driver

Definition

Driver

Operating System aur Hardware ke beech Bridge.

Example

Printer Driver

Wi-Fi Driver

Graphics Driver

Bluetooth Driver

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6. Languages

Machine Language

↓

Assembly Language

↓

High Level Language

Machine Language

0

1

High Level

Python

Java

C++

C

SQL

Assembly

Machine se Easy.

Python se Difficult.

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7. Translators

Compiler

Interpreter

Assembler

Compiler

Pure Program ko

Ek saath Translate.

Interpreter

Line by Line

Translate

Aur Execute.

Assembler

Assembly

↓

Machine Language

---

8. Compiler vs Interpreter

Compiler	Interpreter
Pura Program	Line by Line
Fast Execution	Slow Execution
Error baad me	Error turant
EXE bana sakta hai	Interpreter ke through chalata hai

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9. Personal Computer

Examples

Desktop

Laptop

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10. Processor

Examples

Intel

AMD

Core i3

Core i5

Core i7

Processor

↓

Computer ka Brain.

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11. Internet

Definition

Duniya bhar ke Computers ka Network.

Uses

Communication

Education

Shopping

Banking

Entertainment

Research

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12. E-Commerce

Internet ke through

Buying

Selling

Examples

Amazon

Flipkart

Meesho

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13. Search Engine

Examples

Google

Yahoo

Bing

DuckDuckGo

Kaam

Internet par Information dhoondhna.

---

14. Keyword

Example

Python List Tutorial in Hindi

Ye Search Keyword hai.

🧠 **Memory Tricks (Exam ke liye)**

Computer Types

A

D

H

A = Analog

D = Digital

H = Hybrid

CPU

CU

ALU

RAM

Translator

CIA

C

I

A

Compiler

Interpreter

Assembler

Network

LAN

↓

WLAN

↓

WAN

1. Introduction to Computer

Computer ek **electronic machine** hai jo **Input** leta hai, usko **Process** karta hai, **Storage** me save karta hai aur **Output** deta hai.

Simple Formula

Input → Process → Output → Storage

Real Life Example

Tum Calculator me **2 + 3** enter karte ho.

Input → 2 + 3

↓

CPU Process

↓

Output → 5

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2. Uses of Computer

Computer ka use hota hai:

- 🏠 Home
- 🏦 Bank
- 🏥 Hospital
- 🚆 Railway

- 🎓 Education
- 🏭 Industries
- 🌦️ Weather Forecast
- 🛒 Online Shopping
- 💰 Online Banking
- 🎮 Gaming
- 🤖 AI & Machine Learning

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3. Types of Computer

(A) Analog Computer

- Continuous Data par kaam karta hai.
- Approximate Result deta hai.

Examples

- Thermometer
- Analog Speedometer
- Voltmeter

(B) Digital Computer

- 0 aur 1 par kaam karta hai.
- Accurate Result deta hai.

Examples

- Laptop
- Desktop
- Mobile
- Calculator

(C) Hybrid Computer

Analog + Digital ka combination.

Examples

- Hospital Monitoring Machine
- Weather Forecast System

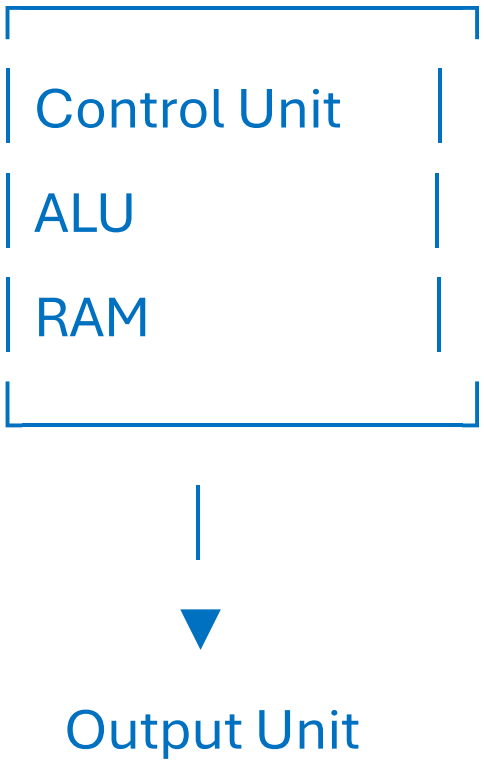
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4. Components of Digital Computer

Input Unit



CPU



Input Unit

User se data leta hai.

Examples

- Keyboard
- Mouse
- Scanner
- Microphone

CPU (Central Processing Unit)

Computer ka Brain.

CPU ke 3 Parts:

1. ALU

Kaam:

- Addition
- Subtraction
- Multiplication
- Division
- Comparison

---

2. Control Unit

Sab devices ko control karta hai.

---

3. RAM

Temporary Memory.

Computer OFF

↓

Data Delete.

Output Unit

Result dikhata hai.

Examples

- Monitor
- Printer
- Speaker

Secondary Memory

Permanent Storage.

Examples

- Hard Disk
- SSD
- Pen Drive
- DVD

---

5. Hardware

Definition

Computer ke Physical Parts ko Hardware kehte hain.

Examples

- Keyboard
- Mouse
- Monitor
- CPU Cabinet
- Printer

---

6. Software

Definition

Programs jo Computer ko kaam karne ki instructions dete hain.

Examples

- Windows
- Python
- MS Word
- Chrome
- VS Code

Difference

Hardware	Software
Physical Part	Program
Touch kar sakte hain	Touch nahi kar sakte
Damage ho sakta hai	Delete/Corrupt ho sakta hai

---

7. Operating System (OS)

Definition

Operating System ek System Software hai jo User aur Hardware ke beech Interface ka kaam karta hai.

Examples

- Windows
- Linux
- macOS
- Android

Functions

✓ Memory Management

✓ File Management

✓ Device Management

✓ Security

✓ Process Management

✓ User Interface

---

8. Driver

Definition

Driver ek Software hai jo Hardware aur Operating System ke beech communication karwata hai.

Examples

- Printer Driver
- Wi-Fi Driver
- Graphics Driver

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9. Languages of Computer

Machine Language

- 0 aur 1
- Computer Directly samajhta hai.

Assembly Language

Machine Language se easy.

Assembler ki help se Machine Language me convert hoti hai.

High Level Language

Human Friendly.

Examples

- Python
- Java
- C
- C++
- SQL

---

10. Translators

Teen Type ke hote hain.

Compiler

Interpreter

Assembler

Compiler

- Pura Program ek saath Translate karta hai.

Interpreter

- Line by Line Translate karta hai.
- Python Interpreter based Language hai.

Assembler

Assembly Language ko Machine Language me convert karta hai.

---

11. Personal Computer (PC)

Personal Use ke liye bana Computer.

Examples

- Desktop
- Laptop

---

12. Processor

Processor Computer ka Brain hai.

Famous Companies

- Intel
- AMD

Famous Series

- Core i3
- Core i5
- Core i7

---

13. Internet

Internet duniya bhar ke Computers ka bahut bada Network hai.

Uses

- Education
- Banking

- Shopping
- Communication
- Entertainment

---

14. E-Commerce

Internet ke through Buying aur Selling.

Examples

- Amazon
- Flipkart
- Myntra

---

15. Search Engine

Internet par Information dhoondhne ka Tool.

Examples

- Google
- Bing
- Yahoo

---

16. Keyword

Search Box me likha gaya Important Word ya Phrase.

Example

Python Tutorial for Beginners

⭐ Memory Tricks

Types of Computer

A D H

A = Analog

D = Digital

H = Hybrid

CPU

CU

ALU

RAM

Translator

CIA

C = Compiler

I = Interpreter

A = Assembler

Network

LAN

↓

WLAN

↓

WAN

Part B: Important Definitions (Exam Ready)

1. Computer

Computer is an electronic device that accepts input, processes data, stores information, and produces output.

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2. Analog Computer

An Analog Computer processes continuous data and gives approximate results.

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3. Digital Computer

A Digital Computer processes binary data (0 and 1) and gives accurate results.

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4. Hybrid Computer

A Hybrid Computer combines the features of Analog and Digital Computers.

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5. CPU

CPU (Central Processing Unit) is the brain of the computer that controls and processes all operations.

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6. ALU

ALU (Arithmetic Logic Unit) performs arithmetic and logical operations.

---

7. RAM

RAM (Random Access Memory) is a temporary memory used to store data while the computer is running.

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8. Hardware

Hardware refers to the physical components of a computer that can be seen and touched.

---

9. Software

Software is a collection of programs that instructs the computer to perform tasks.

10. Operating System

An Operating System is system software that acts as an interface between the user and the hardware.

---

11. Driver

A Driver is software that enables communication between the operating system and hardware devices.

---

12. Machine Language

Machine Language is the binary language (0 and 1) understood directly by the computer.

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13. High-Level Language

A High-Level Language is a programming language that is easy for humans to read and write.

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14. Compiler

A Compiler translates the entire source program into machine code before execution.

15. Interpreter

An Interpreter translates and executes the program one line at a time.

---

16. Assembler

An Assembler converts Assembly Language into Machine Language.

---

17. Personal Computer

A Personal Computer (PC) is a computer designed for individual use.

---

18. Processor

A Processor is the main processing unit that executes instructions and controls the computer.

---

19. Internet

The Internet is a global network that connects millions of computers worldwide.

---

20. E-Commerce

E-Commerce is the buying and selling of goods and services through the Internet.

---

21. Search Engine

A Search Engine is a software tool used to search information on the Internet.

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22. Keyword

A Keyword is a word or phrase entered into a search engine to find relevant information.